03/03/2026: Itô Integration

CSCI 546: Diffusion Models

Textbook reference: Chang Sec 6.9-6.11, GS Sec 13.8-13.9

Announcement (Sign-in Sheet)

Please sign the sign-in sheet.

Announcement (Office Hours Before Midterm)

- Today, 12:15pm-1:15pm, Barnard 352 **or** Wilson.
- Tomorrow (Extra), 2pm-4pm Barnard 352.

Midterm Study Guide

The midterm study guide is now complete, including all questions to study for the midterm.

Review Problem Set #13

Concepts for Problem Set #14

Review: Itô SDEs

Definition. A SDE for the diffusion process X with drift coefficient $\mu(x, t)$ and diffusion coefficient $\sigma(x, t)$ is

$$dX(t) = \underbrace{\mu(X, t)\, dt}_{\text{drift term}} + \underbrace{\sigma(X, t)\, dW(t)}_{\text{diffusion term}}$$

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Observations.

1. *If the noise in the system is due to a multitude of independent random changes, then the Central Limit Theorem predicts the net result will have the normal distribution, a property shared by the increments $W(t) - W(s)$ of the Wiener process.*

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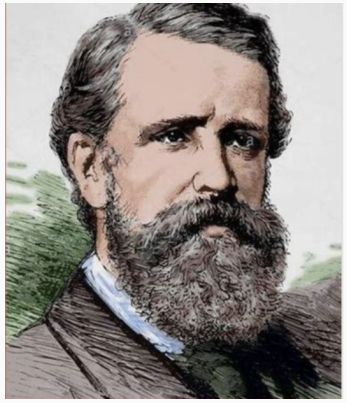
2. Our definition implicitly restricts to Itô SDEs; there are also more general (non-Itô) SDEs.

Itô SDEs: Historical Example

History. In 1900, Louis Bachelier wrote a ground-breaking paper modeling asset prices at the Paris Stock Exchange.



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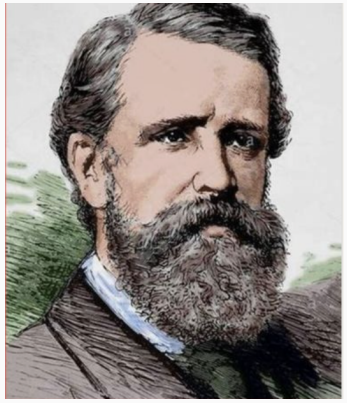

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where σ is a positive constant.

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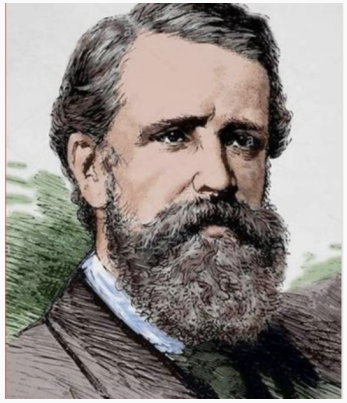

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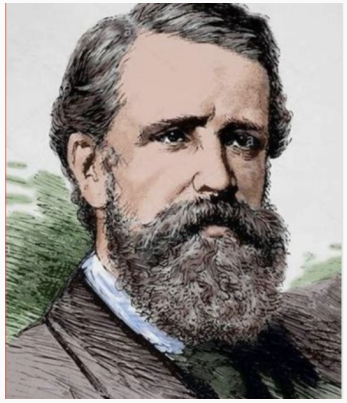

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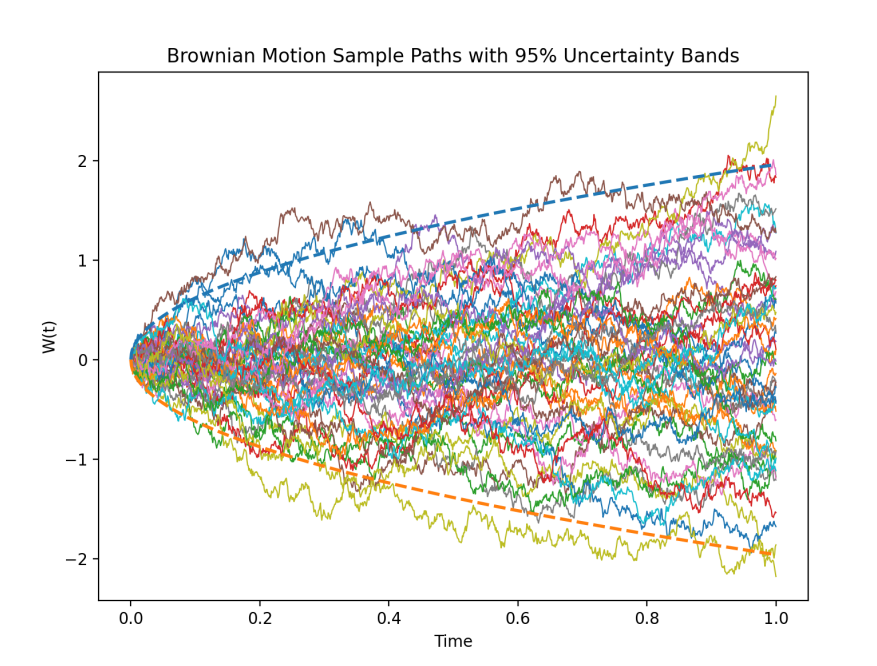

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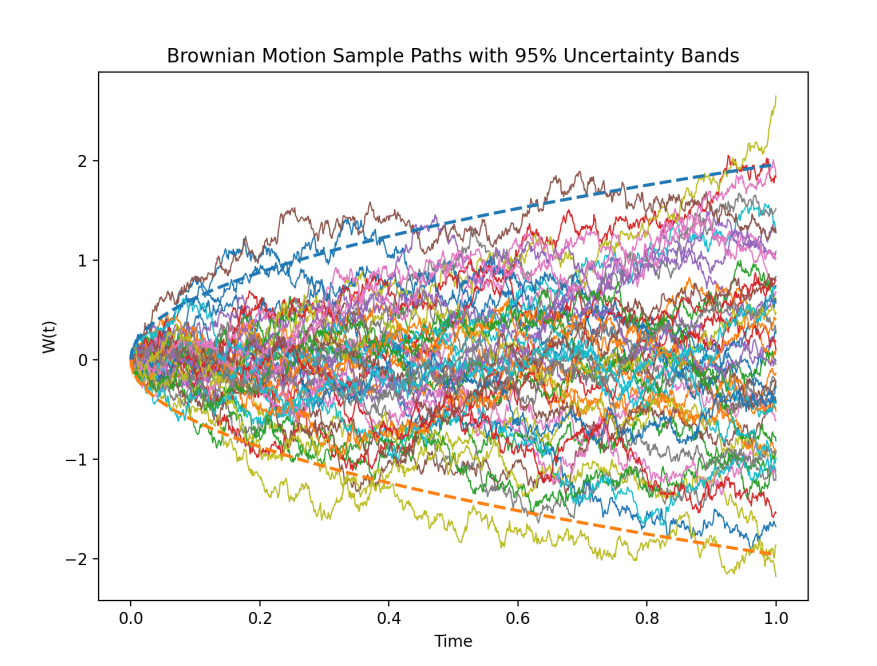

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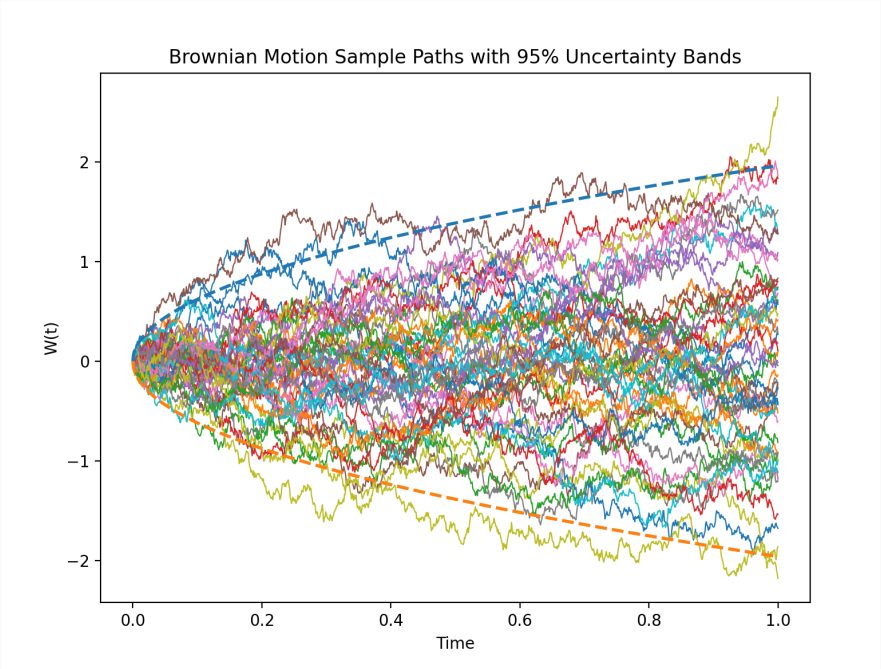

Observation. In this model, stock prices can go negative.

In fact, $P(X_t < 0) = \frac{1}{2}$ at any time t.

Itô SDE Example: Geometric Brownian Motion

A better approach is to have the potential gain or loss $dX(t)$ grow proportionally to the invested sum $X(t)$:

$$\frac{dX(t)}{X(t)} = \sigma \, dW(t)$$

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This is known as `geometric Brownian motion.`

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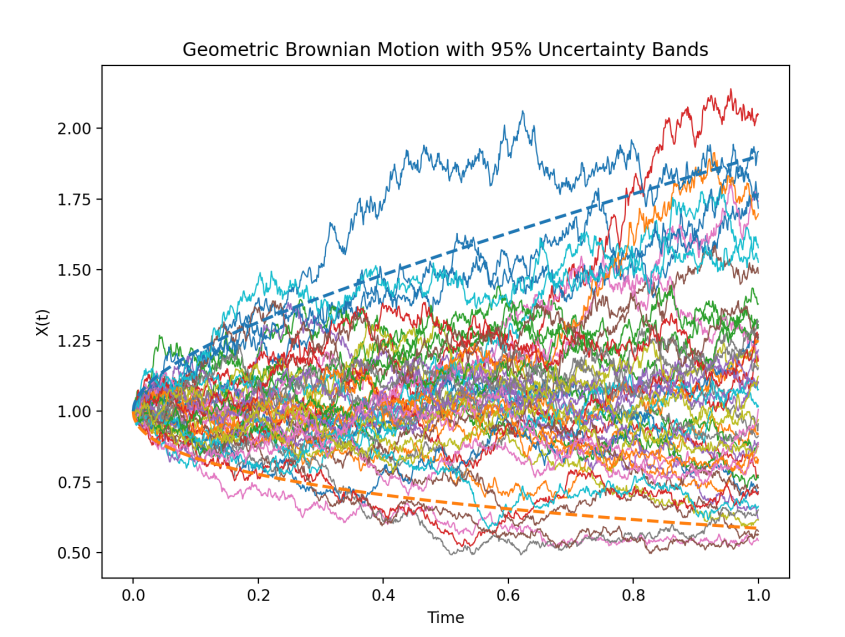
$$dX = \mu X\, dt + \sigma X\, dW$$



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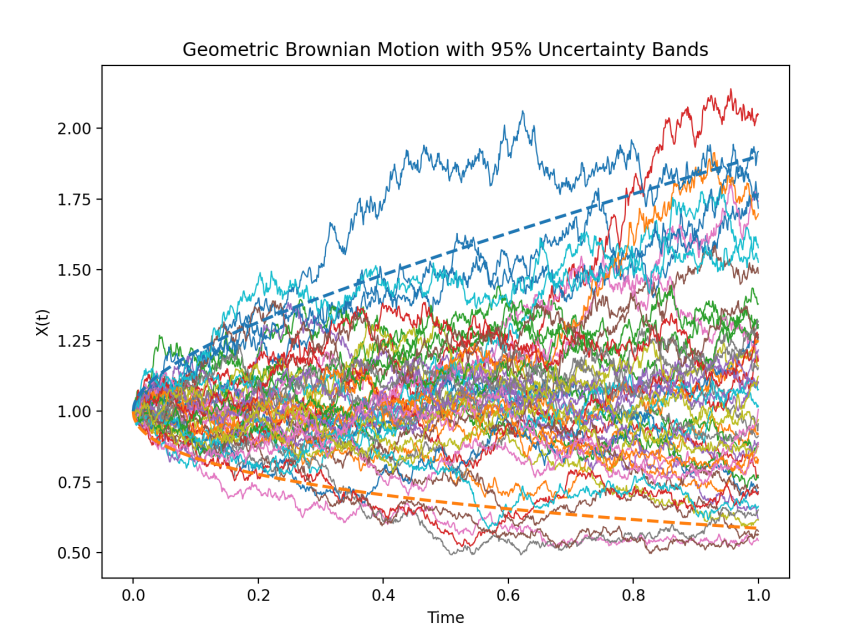

Now:

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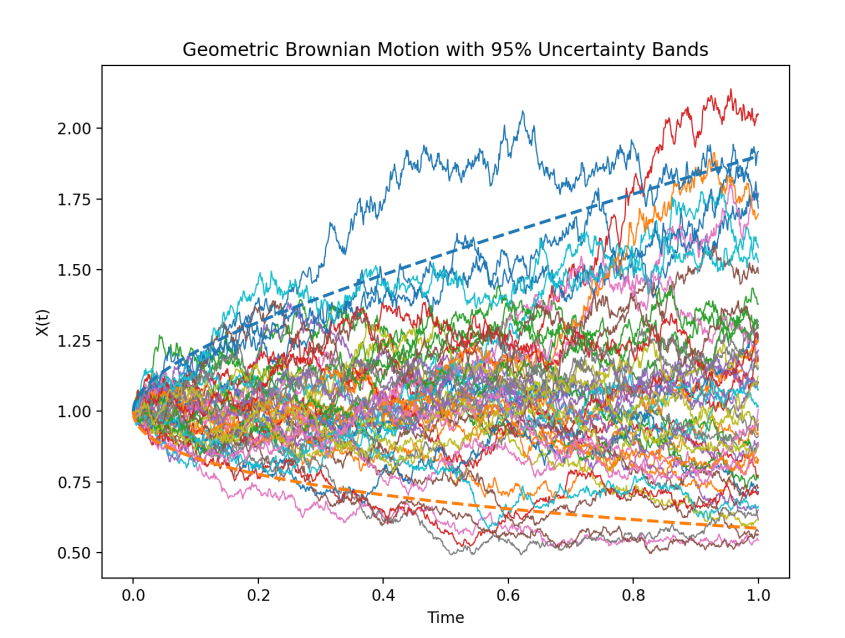

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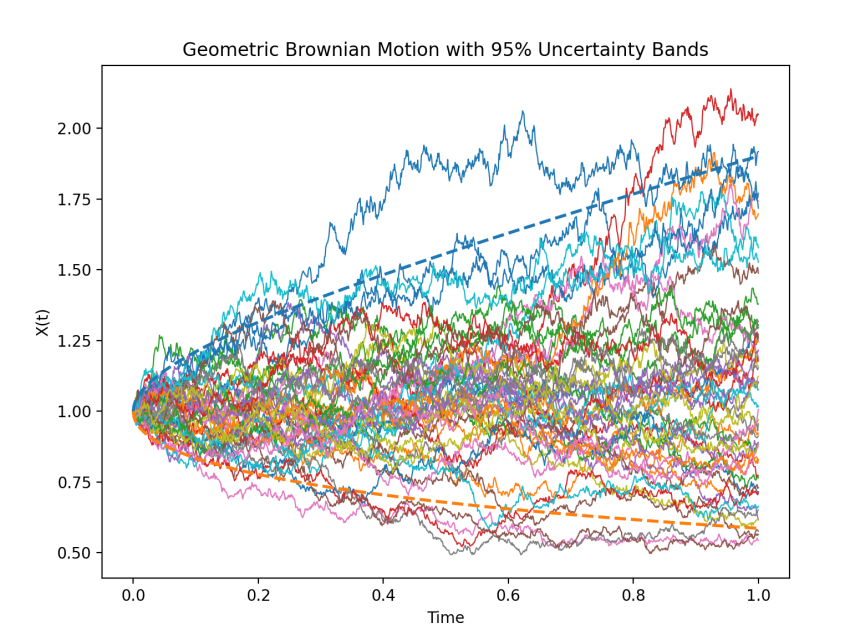

Now:

- The sample paths are always positive.
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- The confidence bands widen exponentially over time.

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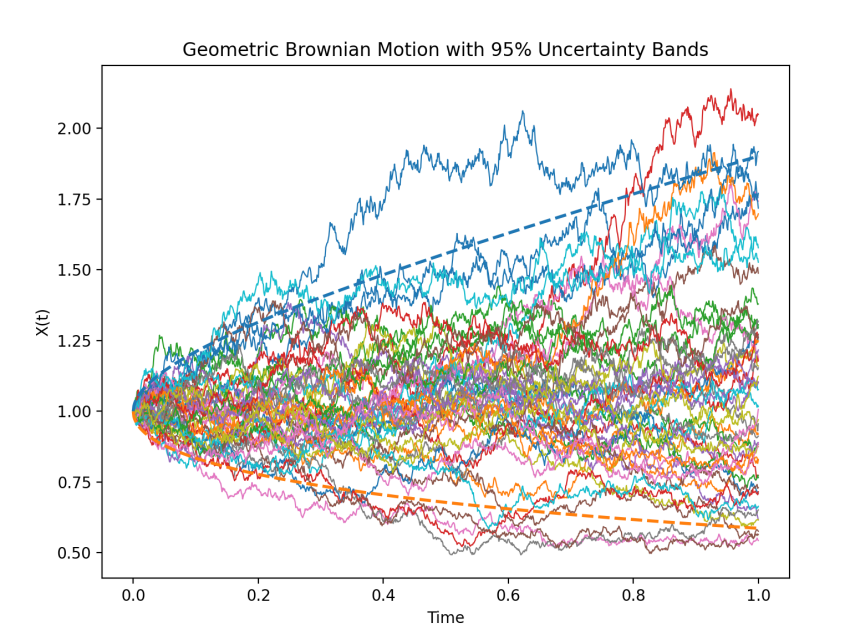

Yet another way to describe geometric Brownian motion is

$$Y_t = e^{X_t},$$

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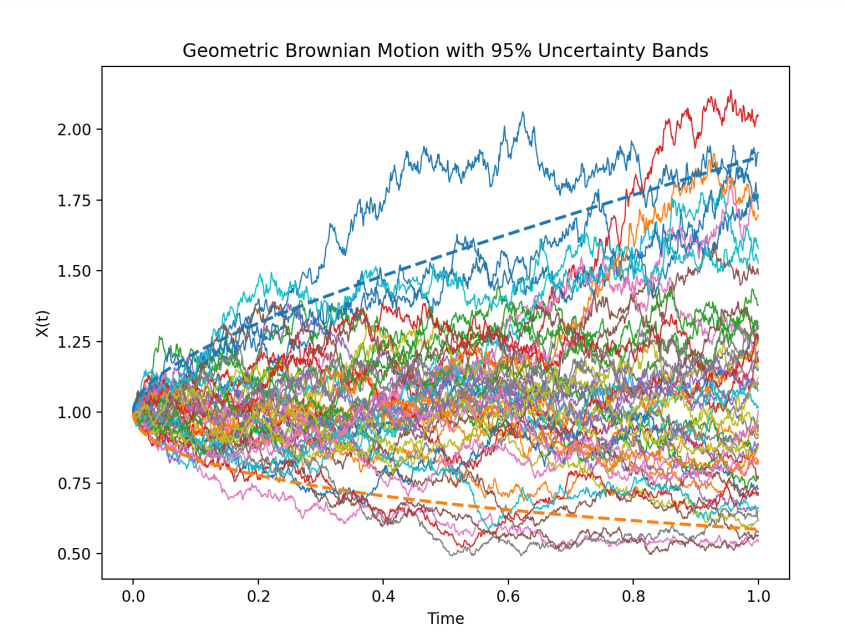

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The desire to compose stochastic processes motivates the Itô chain rule, a chain rule for stochastic processes.

Itô chain rule

Let $X(\cdot)$ be a 1-dimensional real-valued stochastic process which solves the following SDE, constructed from 1-dimensional Brownian motion W

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Intuitively, we would guess by chain rule that

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However, Eq. (2) is wrong! ✗ . Recall that Brownian motion is so irregular that its fluctuations scale with the *square root* of time increments:

$$\boxed{dW \approx (dt)^{\frac{1}{2}}.} \tag{3}$$

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The irregularity in Eq. (3) means that we need to keep additional terms in the Taylor series expansion of the differential:

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  Poll. Where does this term come from?

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Poll. Can anybody explain?

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Reference for this slide: Evans (2012). *An introduction to stochastic differential equations.* For a more technically precise statement of Itô's Lemma, see Evans or Section 5 of Gautam Iyer's *Stochastic Integration. (Lecture notes.)*

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- *Mental health.* $V(T)$ is the emotional dysregulation (or relapse risk accumulation) when the effect of new stress dW_t depends on how stressed W_t the person already is.
  - *Multiplicative risk accumulation.* In general, this can be seen as the simplest mathematical prototype for *risk accumulation in a system where shocks matter more when the system is already strained.*

Problem. How in the world do we compute this integral?!

Itô chain rule: A simple application

In the group exercises, we'll use Itô's chain rule to verify the identity

$$\int_0^T W\,dW = \frac{1}{2}\Big(W^2(T) - T\Big).$$

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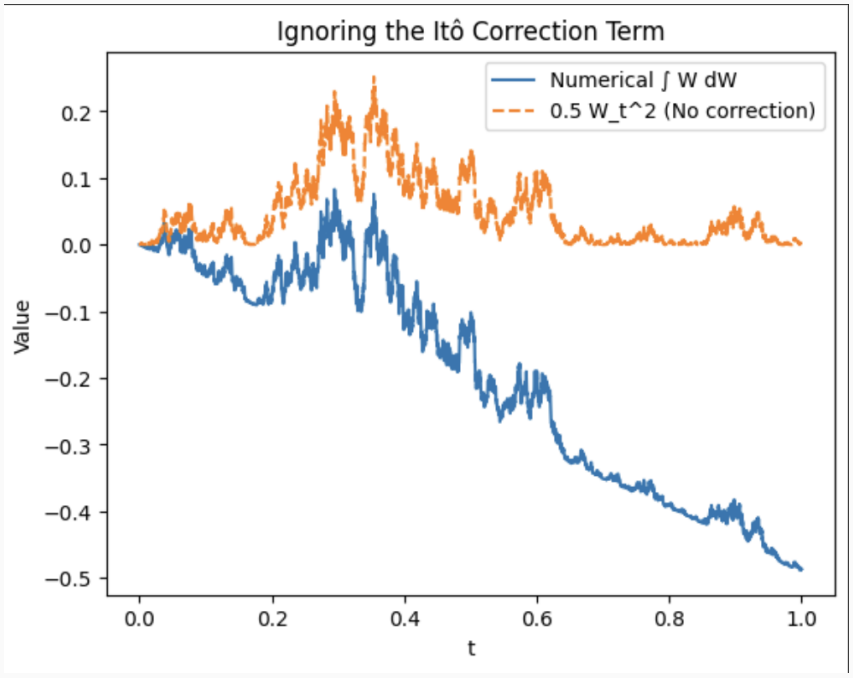
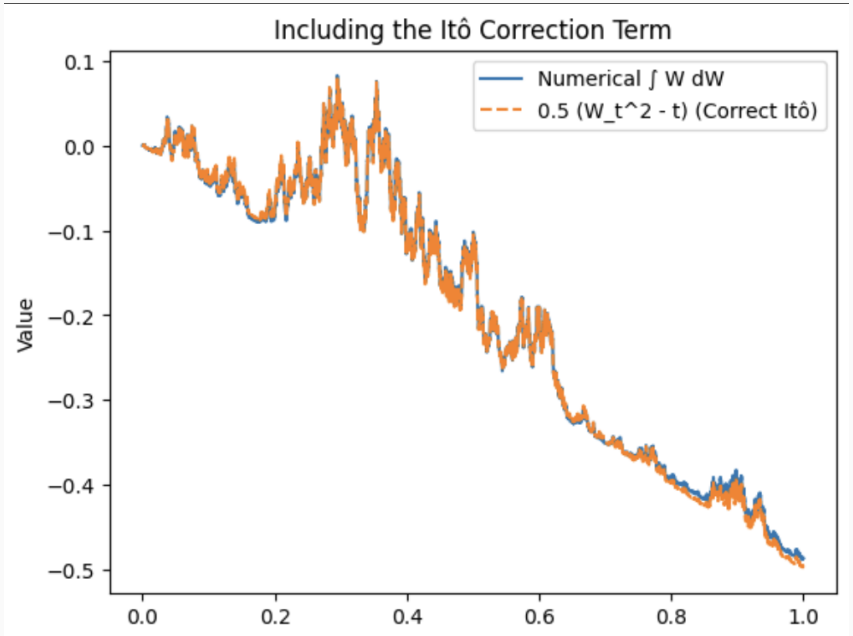
Remarks.

- In stochastic calculus, the result of the definite integral is a random variable, not a *number* as in ordinary calculus.
- Also note the extra term " $-T$ " not present in ordinary calculus.
- Without this "Itô correction term", the integral is not correct, as can be verified by numerical simulations.

Verifying the Itô correction term





Verifying the Itô correction term

Review. Recall that we could interpret the integral

$$V(T) = \int_0^T W_t \, dW_t$$

as the total profit from a betting strategy where a person bets an amount proportional to one's current position/capital W_t.

Observation. This gives another corroboration of Itô's correction term $\int_0^T W_t \, dW_t = \frac{1}{2}\big(W^2(T) - T\big)$: without it, there would be a way to make unlimited money by betting (arbitrage)!

Random Groups

Aubrey Williams: 5
Austin Barton : 7
Blake Sigmundstad: 5
Diego Moylan: 6
Dillon Shaffer: 1
Ismoiljon Muzaffarov: 1
Jacob Tanner: 8
Josh Stoneback: 3
Joshua Bowen: 2
Joshua Culwell: 1
Laura Banaszewski: 2
Lina Hammel: 6
Logan Racz: 2
Matt Hall: 8
Mike Kadoshnikov: 3
Owen Cool: 4
Racquel Bowen: 4
Samuel Mocabee: 3
Tatiana Kirillova: 7

Group exercises - Problem Set 14

The following exercises are about Itô's chain rule (also called Itô's Lemma).

1. (Chang Sec 6.9) **Geometric Brownian motion.** Let X_t be a stochastic process governed by the Itô SDE

$$dX_t = \mu dt + \sigma dW_t.$$

   Define a new stochastic process by $Y_t = e^{X_t}$. Use Itô's Lemma to show that Y_t is a geometric Brownian motion, and find the infinitesimal parameters of Y_t with respect to the infinitesimal parameters of X_t. Provide an interpretation of the infinitesimal parameters.

2. (Gautam Iyer, *Stochastic Integration. (Lecture notes.)* , Exercise 6.1.) **Integrating Brownian motion against Brownian motion.** Use Itô's Lemma to verify the identity

$$\int_0^T W\, dW = \frac{1}{2}\Big(W^2(T) - T\Big).$$

   Compute $\mathbb{E}[\int_0^T W\, dW]$.